OBJECT ORIENTED SOFTWARE ENGINEERING

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Pattern Based Design

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- When expert works on a **particular problem**, it is usual for them to **tackle it** by inventing a **new solution** that is completely distinct from existing ones.

- They often **recalls a similar problem** they have already solved and **reuse** the essence of its solution to solve new problem

A brief background

- In 1994, four authors **Erich Gamma, Richard Helm, Ralph Johnson and John Vlissides** published a book titled **Design Patterns - Elements of Reusable Object-Oriented Software** which initiated the concept of **Design Pattern** in Software development.

- These authors are collectively known as **Gang of Four** (**GOF**).

What is a Pattern?

Each pattern is a **three-part** rule, which expresses a relation between

- **a certain context**
- **a problem**
- **a solution.**

- In short, Pattern is a
  - Thing, which happens in the world.
  - A rule, which tells how to create that thing and when we must create it.

- A **design pattern is not a framework** and is not directly deployed via code.

- A design pattern is **not a finished design** that can be transformed directly into **source or machine** code

What makes Pattern?

Three-part schema that underlies every pattern:

Context: **a situation giving rise to a problem.**

Problem: **the recurring problem arising in that context.**

Solution: **a proven resolution of the problem**

Patterns advantages

The advantages of using the design patterns are given below.

1. All design patterns are **language neutral**.
2. Patterns offer programmers to select a **tried and tested solution** for the specific problems.
3. It consists of record of execution to **decrease any technical risk** to the projects.
4. Patterns are **easy to use and highly flexible.**

THANK YOU

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